

HARDIK SRIVASTAVA

hardiksrwork@gmail.com | +1 (206) 773-9806 | Seattle, WA
Portfolio | LinkedIn | LeetCode | ResearchGate | Google Scholar | GitHub

EDUCATION

University of Washington <i>Master of Science in Data Science</i>	Seattle, WA Sept 2025 - Mar 2027
SRM Institute of Science and Technology <i>Bachelor of Technology in Computer Science with specialization in Big Data Analytics (GPA - 3.82/4)</i>	Chennai, TN May 2019 - Jun 2023

WORK EXPERIENCE

Graduate Research Assistant (P.I. - Prof. Zaid Harchaoui) Paul G. Allen School of Computer Science, University of Washington	Sept 2025 - Present <i>Seattle, WA</i>
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- Engineered and fine-tuned **modular LoRA adapters** for **controllable text style transfer**, then built **metric-driven agents** to anonymize text via learned style representations, **constrained decoding** and attribution classifiers under privacy-fidelity constraints.
- Formulated closed-loop control logic to dynamically remix text across **16+ writing styles**, integrating **Mistral** models for automated evaluation. Boosted author anonymity by **40%** on a 30K+ corpus while preserving **88%** semantic similarity and **5%** perplexity drift.
- Optimized agent policies via **multi-objective feedback** and conducted empirical analysis across multilingual datasets. Benchmarked the system against larger LLM baselines, achieving **privacy gains** at **3x lower inference cost** and **34% fewer generated tokens**.

Applied Scientist 2 JPMorgan Chase & Co.	Jun 2023 - Aug 2025 <i>Hyderabad, TS</i>
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- Led **fraud detection** initiatives saving over **\$220M annually** by developing high-recall **XGBoost** and **CatBoost**; optimized recall-precision tradeoffs using **constrained objective tuning** while maintaining feature interpretability for stakeholders.
- Productionized Semi-Supervised Graph Neural Networks across 60M+ transaction nodes; introduced **topology-aware edge sampling** and **dynamic neighborhood aggregation**, boosting fraud-detection **precision** by **9.2%** over **2M transactions/day**.
- Architected a GPU-served **contrastive pre-trained Transformer** (ViT + BERT-style decoder), scaling to **500K+** requests/day, for OCR-based check fraud detection. Improved fraud precision by **19%** while reducing latency via batched mixed-precision inference.
- Built a **Learning-to-Rank** entity matching system over **3B+** KYC records using Pairwise Ranking loss to flag fraud entities; Outperformed the existing TF-IDF baselines, cutting **62% false positives** and saving **700+** analyst review hours daily in production.
- Developed **distributed data pipelines** in PySpark to handle terabyte-scale transactions. Performed **optimal feature selection** using **Decision Trees** that boosted **Fraud Capture Rate** by **14%** and enhanced predictive accuracy for risk scoring models.

Research Intern McGill University - <i>Mitacs Scholar</i>	Jun 2022 - Sept 2022 <i>Montreal, QC</i>
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- Developed **RoBERTa-based sentence embeddings** on AAC image captions to capture word-context. Achieved **Krippendorff's** alpha reliability score of **0.82** for context-aware vocab retrieval; deployed the model in **ClickAAC** mobile application.

Machine Learning Engineer - Intern Samsung Research	Jan 2022 - Jun 2022 <i>Bangalore, KN</i>
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- Optimized **word-context relationships** for Samsung Bixby search using bipartite graphs, reducing **false-positive** preds by **12%**. Matched **FP32** perf for INT8 inference using quantization-aware training for word-disambiguation task with a **2.9%** F1 gain.

PROJECTS

Agentic Ad Intelligence Engine [code] Architected a **multi-agent orchestration** pipeline for X (Twitter) using **Grok 4.1** Reasoning models to generate dynamic user personas and personalized ad copy from real-time behavioral signals. Engineered an LLM-driven routing system that uses ensemble engagement predictions and feedback-driven ranking to optimize context-aware ad delivery and in-feed placement.

KeyCognition: Behavioral AI Platform [code] Engineered a temporal sequence modeling pipeline in Databricks, processing **136M+** real-time **keystroke** events to construct latent behavioral embeddings for **cognitive load prediction**. Developed a hybrid behavioral classification model fusing LSTMs and K-Means clustering to extract micro-temporal typing anomalies like backspaces, flight times etc.

PUBLICATIONS

- [Srivastava, H.](#) (2023). Multi-Modal Sentiment Analysis using Text and Audio for Customer Support Centers
- [Srivastava, H.](#) (2023). Neural Text Style Transfer with Custom Language Styles for Personalized Communication Systems
- [Srivastava, H.](#) (2021). Using NLP Techniques for Enhancing Augmentative and Alternative Communication Applications

SKILLS

AI/Research	Machine Learning, Natural Language Processing, Generative AI, Graph-ML, Transformers, Ranking, Agentic AI Large Language Models, Time Series Forecasting, Data Analysis
CS/Tooling	C++, Python, Java, SQL, PyTorch, TensorFlow, Sklearn, Spark, Kafka, W&B, HuggingFace, Optuna, Git
Infra/MLOps	AWS, Docker, Kubernetes, Flask, MLFlow, RAG, LangChain, NVIDIA Triton, Feature Stores, Jenkins